

Summary:

Results-driven **Data Scientist** with 4+ years of proven success delivering **\$15M+ in business value** across **e-commerce, retail, and financial services**. Expert in designing and deploying **end-to-end machine learning** and **generative AI** solutions impacting **2M+ global users**, with measurable gains in **conversion rates, revenue growth, and operational efficiency**. Skilled in **predictive analytics, causal inference, A/B testing, and real-time experimentation platforms**. Proficient in **Python, R, SQL**, and **advanced ML frameworks** including **Scikit-learn, XGBoost, Random Forest, Transformers, LangChain, HuggingFace**, and **LLM fine-tuning**. Strong experience in **RAG pipelines, vector databases (FAISS, Pinecone), MLOps**, and **cloud-native deployments** on **AWS (S3, Lambda, SageMaker), Azure, GCP, Snowflake**, and **Databricks**. Adept at building **scalable data pipelines** and **interactive dashboards** with **Tableau, Power BI, and Streamlit** to drive **data-informed decision-making**. Passionate about leveraging **cutting-edge AI advancements** such as **Generative AI, RAG optimization, hybrid search architectures, and causal ML** to solve complex business challenges and deliver sustainable competitive advantage.

Skills:

- **Programming Languages:** Python, R, SQL, C, C++, Node.js
 - **Machine Learning & AI:** Regression, Classification, Clustering, Time-Series Forecasting, XGBoost, Random Forest, Gradient Boosting, Neural Networks (CNN, RNN, LSTM), Transformers, Generative AI, RAG, Prompt Engineering.
 - **Generative AI & NLP:** LangChain, HuggingFace, OpenAI API, SBERT, Vector Databases (FAISS, Pinecone), RAG Optimization, Hybrid Search, LLM Fine-Tuning
 - **Data Analysis:** Pandas, NumPy, Scikit-learn, RStudio, Spark, Seaborn, Matplotlib, Google Analytics
 - **Experimentation & Causal Inference:** A/B Testing, CUPED, CUPAC, Bayesian Optimization, Propensity Score Matching (PSM), Statistical Analysis
 - **Cloud Platforms:** AWS (S3, Lambda, SageMaker), Azure, GCP, Snowflake, Databricks
 - **MLOps & Data Engineering:** Data Pipeline Development, Feature Engineering, Model Deployment, Model Monitoring, Data Version Control (DVC), CI/CD for ML
 - **Databases:** MySQL, PostgreSQL, MongoDB, Vector DB
 - **Visualization & BI Tools:** Tableau, Power BI, Streamlit, Matplotlib, Seaborn
 - **Other Tools:** Git, Docker, Kubernetes, Jira, Confluence
-

Education:

- **Master's of Science in Data Science** From New Jersey Institute of Technology, **New Jersey, USA**
-

Professional Experience

Capital One | Data Scientist | Jan 2024 – Present

Project Description: Worked on multiple high-impact projects aimed at optimizing e-commerce performance and financial insights for small and medium-sized enterprises. Contributed to the development and evaluation of ML-driven recommendation systems, A/B testing pipelines, and lending analysis platforms. Built and deployed scalable data pipelines and Generative AI solutions, including a Retrieval-Augmented Generation (RAG) system to automate internal HR query resolution. Integrated advanced techniques like Propensity Score Matching and CUPED to support decision-making through causal inference and experimentation analytics.

- Achieved **\$250K operational cost savings** by monitoring CTR, Conversion Rate, and AOV during a 6-week **A/B test** of an **ML-powered recommendation system**, deploying models via **AWS SageMaker** to enable data-driven e-commerce decisions.
- Reduced experiment timelines by **30%** by implementing **CUPED with logistic regression**, lowering variance in conversion rate estimates by 13% for faster, statistically reliable outcomes.
- Uncovered **7.6% SME revenue growth opportunities** by applying **Propensity Score Matching (PSM)** and **causal inference** to assess the impact of instant lending, influencing strategic investment decisions.
- Designed and automated **scalable data pipelines** using **AWS and Databricks**, streamlining ingestion, cleaning, and feature engineering for 5,000+ SME records, cutting preprocessing time by **40%** and enabling **real-time analytics**.
- Enhanced HR operations by **40% efficiency** through a **Retrieval-Augmented Generation (RAG)** pipeline integrated with an **LLM chatbot**, enabling multi-turn dialogue and human-in-the-loop feedback for accurate internal query resolution.
- Built an advanced **hybrid RAG architecture** leveraging **SBERT + FAISS Vector DB**, custom reranking (cosine similarity, token confidence), and **reciprocal rank fusion**, optimizing 1,200+ documents for high-relevance, context-aware responses.

EVALUESERVE | Data Scientist | Aug 2019 – Jul 2022

Project Description: Engaged in AI/ML projects across the banking and retail sectors, designing predictive models and NLP-based automation solutions. Developed classification and segmentation systems to optimize campaign performance and operational workflows. Applied deep learning architectures (CNNs, LSTMs) for text processing in loan verification systems, and utilized feature engineering and advanced regression models to solve high-dimensional business problems. Built real-time dashboards to empower decision-makers with accessible and actionable insights.

- Automated loan verification for HSBC by developing **NLP-based text extraction and classification models** using **CNNs, RNNs, and attention-based LSTMs**, reducing document processing time by **60%** and increasing loan completion rates by **25%**.
- Improved campaign ROI by **15%** for HSBC through a **Customer Spend Prediction model** built with **Python, SQL, and ML algorithms**, processing 2B+ transactions for 25M customers and implementing a model drift monitoring system.
- Achieved an **adjusted R² of 0.87** in predictive models by engineering advanced features, reducing multicollinearity via **VIF**, and leveraging **Random Forest, XGBoost, and GBM** to handle class imbalance and high-dimensional datasets.
- Reduced ERP/CRM operational costs by **\$20K annually** and doubled efficiency for PVH during the pandemic by creating a **retail planning dashboard** with **Python, SQL, and Tableau** for demand forecasting and inventory optimization.
- Increased supplier feedback by **45%** and topline revenue by **3%** for Belk by enhancing **K-Means clustering** with weighted feature selection, dynamic cluster sizing, and advanced statistical analysis for business-aligned customer segmentation.